

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Cryptology – Lab_8

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Write a python script to implement RSA algorithm

main.py

```python
from math import gcd

p = int(input("Enter prime p: "))
q = int(input("Enter prime q: "))
e = int(input("Enter e: "))
m = int(input("Enter message (number): "))

n = p * q
phi = (p - 1) * (q - 1)

if gcd(e, phi) != 1:
    print("Invalid e! gcd(e, phi) must be 1")
    exit()
if m >= n:
    print("Message must be less than", n)
    exit()

d = pow(e, -1, phi)       # private key
c = pow(m, e, n)          # encrypt
r = pow(c, d, n)          # decrypt

print("Public key :", (e, n))
print("Private key:", (d, n))
print("Encrypted  :", c)
print("Decrypted  :", r)
```

Output:

Enter prime p: 61

Enter prime q: 53

Enter e: 17

Enter message (number): 65

Public key : (17, 3233)

Private key: (2753, 3233)

Encrypted : 2790

Decrypted : 65

Write a python script to implement DH Key exchange

main.py

```python
p = int(input("Enter prime p: "))
g = int(input("Enter base g: "))
a = int(input("Alice's secret a: "))
b = int(input("Bob's secret b: "))

A = pow(g, a, p)          # Alice sends A
B = pow(g, b, p)          # Bob sends B

key_alice = pow(B, a, p)
key_bob = pow(A, b, p)

print("A =", A, " B =", B)
print("Alice's key:", key_alice)
print("Bob's key  :", key_bob)
print("Keys match :", key_alice == key_bob)
```

Output:

Enter prime p: 23

Enter base g: 5

Alice's secret a: 6

Bob's secret b: 15

A = 8

B = 19

Alice's key: 2

Bob's key : 2

Keys match : True